

Zeotap Assignment Submission

1. Personal Details

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Position: Software Engineer Intern

GitHub Repo: <https://github.com/dakshrana205/wrangler>

2. Assignment Summary

I enhanced the CDAP Wrangler tool by adding support for Byte Size (e.g., KB, MB) and Time Duration (e.g., ms, s). These units are often used in logs and performance data. To demonstrate their use, I also created a new directive called 'aggregate-stats', which can calculate totals or averages using these units. This makes data processing faster and easier.

3. Work Completed

- Grammar Update:

Added lexer rules for BYTE_SIZE and TIME_DURATION with helper unit fragments.

- API Updates (wrangler-api):

Created ByteSize.java and TimeDuration.java classes.

- Core Updates:

Added parser visit methods in RecipeVisitor.java.

- New Directive:

Implemented AggregateStats.java for aggregation.

- Unit Tests:

ByteSizeTest.java

TimeDurationTest.java

AggregateStatsTest.java

4. Sample Usage

Recipe:

aggregate-stats :data_transfer_size :response_time total_size_mb total_time_sec

Input:

data_transfer_size,response_time

1KB,100ms

2MB,500ms

500KB,200ms

Output:

total_size_mb,total_time_sec

2.5,0.8

5. AI Tool Prompts Used

1. Add lexer rules for BYTE_SIZE and TIME_DURATION in Directives.g4
2. Implement ByteSize and TimeDuration classes extending Token
3. Add visit methods for byteSizeArg and timeDurationArg in RecipeVisitor
4. Implement AggregateStats directive
5. Write unit tests for all new features
6. Verify build and test execution

6. Build & Test Results

```
PROBLEMS 1K+ OUTPUT DEBUG CONSOLE TERMINAL PORTS
[INFO] --- resources:3.3.1:resources (default-resources) @ wrangler-transform ---
[INFO] skip non existing resourceDirectory /Users/dakshrana/Desktop/wrangler/wrangler-transform/src/main/resources
[INFO] --- compiler:3.13.0:compile (default-compile) @ wrangler-transform ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 5 source files with javac [forked debug target 1.8] to target/classes
[WARNING] [options] bootstrap class path is not set in conjunction with -source 8
[INFO] Reactor Summary for Wrangler 4.12.0-SNAPSHOT:
[INFO]
[INFO] Wrangler ..... SUCCESS [ 0.877 s]
[INFO] Wrangler API ..... SUCCESS [ 2.623 s]
[INFO] Wrangler REST Protocol Classes ..... SUCCESS [ 2.235 s]
[INFO] Wrangler Testing Framework ..... SUCCESS [ 2.190 s]
[INFO] Wrangler Core ..... SUCCESS [ 9.071 s]
[INFO] Wrangler Storage ..... SUCCESS [ 1.620 s]
[INFO] Wrangler Service ..... SUCCESS [ 2.826 s]
[INFO] Wrangler Transform ..... SUCCESS [ 0.983 s]
[INFO] BUILD SUCCESS
[INFO] Total time: 23.281 s
[INFO] Finished at: 2025-04-13T01:08:25+05:30
[INFO]
dakshrana@192 wrangler %
```

```
[INFO] Reactor Summary for Wrangler 4.12.0-SNAPSHOT:
[INFO]
[INFO] Wrangler ..... SUCCESS [ 1.580 s]
[INFO] Wrangler API ..... SUCCESS [ 2.376 s]
[INFO] Wrangler REST Protocol Classes ..... SUCCESS [ 4.119 s]
[INFO] Wrangler Testing Framework ..... SUCCESS [ 2.696 s]
[INFO] Wrangler Core ..... SUCCESS [ 36.540 s]
[INFO] Wrangler Storage ..... SUCCESS [ 2.580 s]
[INFO] Wrangler Service ..... SUCCESS [ 7.726 s]
[INFO] Wrangler Transform ..... SUCCESS [ 2.801 s]
[INFO] BUILD SUCCESS
[INFO] Total time: 01:01 min
[INFO] Finished at: 2025-04-13T01:16:56+05:30
[INFO]
dakshrana@192 wrangler %
```

```
PROBLEMS 757 OUTPUT DEBUG CONSOLE TERMINAL PORTS
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.proto, has 1, private references [io.cdap.cdap.api],
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.proto.connection, has 1, private references [javax.annota
tion],
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.proto.gcs, has 1, private references [javax.annotation],
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.proto.recipe.v2, has 1, private references [io.cdap.cdap
.api],
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.proto.schema, has 1, private references [javax.annotatio
n],
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.proto.workspace, has 1, private references [javax.annota
tion],
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.proto.workspace.v2, has 2, private references [io.cdap.c
dap.api, javax.annotation],
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.registry, has 3, private references [io.cdap.cdap.api.se
rvice.http, javax.annotation, io.cdap.cdap.api.service.worker],
[WARNING] Bundle io.cdap.wrangler:wrangler-transform:jar:4.12.0-SNAPSHOT : Export io.cdap.wrangler.utils, has 3, private references [org.apache.avro, javax
.annotation, org.apache.commons.collections4],
[INFO] --- install:3.1.2:install (default-install) @ wrangler-transform ---
[INFO] Installing /Users/dakshrana/Desktop/wrangler/wrangler-transform/pom.xml to /Users/dakshrana/.m2/repository/io/cdap/wrangler/wrangler-transform/4.12.0-
SNAPSHOT/wrangler-transform-4.12.0-SNAPSHOT.pom
[INFO] Installing /Users/dakshrana/Desktop/wrangler/wrangler-transform/target/wrangler-transform-4.12.0-SNAPSHOT.jar to /Users/dakshrana/.m2/repository/io/cd
ap/wrangler/wrangler-transform/4.12.0-SNAPSHOT/wrangler-transform-4.12.0-SNAPSHOT.jar
[INFO] Reactor Summary for Wrangler 4.12.0-SNAPSHOT:
[INFO]
[INFO] Wrangler ..... SUCCESS [ 1.862 s]
[INFO] Wrangler API ..... SUCCESS [ 5.440 s]
[INFO] Wrangler REST Protocol Classes ..... SUCCESS [ 4.123 s]
[INFO] Wrangler Testing Framework ..... SUCCESS [ 1.663 s]
[INFO] Wrangler Core ..... SUCCESS [ 44.698 s]
[INFO] Wrangler Storage ..... SUCCESS [ 3.688 s]
[INFO] Wrangler Service ..... SUCCESS [ 21.980 s]
[INFO] Wrangler Transform ..... SUCCESS [ 10.337 s]
[INFO] BUILD SUCCESS
[INFO] Total time: 01:34 min
[INFO] Finished at: 2025-04-13T01:28:21+05:30
[INFO]
dakshrana@192 wrangler %
```

All tests passed successfully:

- Wrangler API: 2 tests passed
- Wrangler Proto: 3 tests passed
- Wrangler Core: 367 tests passed (2 skipped)
- Wrangler Storage: 5 tests passed
- Wrangler Service: 17 tests passed (6 skipped)
- Wrangler Transform: 2 tests passed

Total: 396 tests passed, 8 skipped, 0 failures

'mvn clean install' and 'mvn test' ran successfully with no checkstyle errors.